- Because of their large numbers other species do not have space in the intestine
- Acidic environment created by vaginal Lactobacilli suppresses growth of other bacteria.

HOTS

How do normal flora help host against pathogenic microorganisms?

Exogenous sources of infections

Human beings: The commonest source of infections in human are from other human beings. The parasite may originate from a patient or a carrier. A patient is a person who harbours the pathogenic microorganism and suffers from ill effect because of it. A **healthy carrier** is the one who harbours the pathogens but has never suffered from the disease caused by the pathogen. A **convalescent carrier** is one who has recovered from the disease and continues to harbor the pathogen in his body (Figure 12.2).

Animals: Many pathogens are able to infect both human beings and animals. Infectious disease transmitted from animals to human beings are called zoonoses. Zoonotic diseases may be bacterial (Example: plague from rats) or viral (Example: rabies from dogs).

Insects: Blood sucking insects may transmit pathogens to human beings. The diseases so caused are called arthopod borne diseases. Insects such as mosquitoes, ticks, mites, flies, fleas and lice that transmit infections are called vectors.

Transmission may be mechanical or biological. Mechanical transmission is the passive transport of the pathogens on the insects feet or other body parts. Example: Houseflies can transfer the pathogens of Typhoid fever and Bacillary dysentery

Infobits

The story of typhoid Mary

The classic example of role of carriers in disease transmission is the story of Mary Mallon.

Mary Mallon was an Irish immigrant who worked as a cook in New York in the early twentieth century. Over seven years, from 1900 to 1907, Mallon worked for number of different households. Unknowingly spreading illness to the people who lived in each one. Later George Soper, tracked Mallon linked 22 cases of typhoid fever through her. He discovered that Mallon was a carrier for typhoid but was immune to it herself. Although active carriers had been recognized before, this was the first time that an asymptomatic carrier of infected had been identified. Epidemiologists were able to trace 51 cases of typhoid fever and three deaths directly to Mallon, who is remembered as "Typhoid Mary". She was forced to prison and then released under the conditions that she could no longer be a cook. She assumed a false name and began cooking again and of course, infecting numerous people. She was again prisoned where she died 26 years later of pneumonia.

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